Points, Lines, Rays, and Polygons Mission

Warm-up and worked example

Name: ____________________ **Date:** ______________

PAGE 1 | WEEK 8, DAY 1

YOUR MISSION **Describe and draw points, lines, line segments, rays, intersecting lines, parallel lines, perpendicular lines, and polygons.**	**SMART MOVE** **Name the attribute being measured before choosing a method.**

Math Talk

What is the same and different about a line segment and a ray?

My idea:

____

New Words

attribute	parallel	perpendicular
My meaning or picture:	My meaning or picture:	My meaning or picture:

Worked Example

Do this example with your teacher. Say what changes and what stays the same.

1. Draw and label a point.
2. Draw and label a line, segment, and ray.

What the example shows: __________________________________

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Learn it and show your thinking

Name: ______________________ **Date:** _______________

YOUR MISSION **Follow the learning path, then explain the big idea.**	**SMART MOVE** **Draw before asking for a rule.**

[ ] Step 1: Draw and label a point.

[ ] Step 2: Draw and label a line, segment, and ray.

[ ] Step 3: Identify intersecting, parallel, and perpendicular lines.

[ ] Step 4: Build polygons from line segments.

[ ] Step 5: Sort shapes and non-shapes by attributes.

My model, drawing, or organized work:

The big idea in my own words:

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Practice lab

Name: ______________________ **Date:** _______________

PAGE 3 | WEEK 8, DAY 1

YOUR MISSION Use a model, equation, or explanation for every task.	**SMART MOVE** Check whether your answer is reasonable.

1. Draw each geometry object.

2. Identify examples around the room.

3. Classify triangles, quadrilaterals, pentagons, and hexagons.

4. Explain why a curved shape is not a polygon.

Choose one answer and prove it another way:

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Challenge and final check

Name: ______________________ **Date:** _______________

PAGE 4 | WEEK 8, DAY 1

YOUR MISSION Try an unfamiliar problem and defend your reasoning.	**SMART MOVE** A wrong first idea can still lead to a strong solution.

Challenge Mission

Draw a shape with exactly one pair of parallel sides and at least one right angle.

My plan:

My solution and proof:

Final Check

Explain the difference between parallel and perpendicular lines.

[ ] I can teach it	[ ] I need practice	[ ] I need help

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Reusable math tool

Name: ______________________ **Date:** _______________

PAGE 5 | WEEK 8, DAY 1

YOUR MISSION **Use this page to draw, model, organize, and check.**	**SMART MOVE** **Name the attribute being measured before choosing a method.**

Grid Lab

What does your drawing prove?

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